

Abstract

BloodBond application aims seamless communication between blood donors and hospitals, ensuring timely blood donations and distribution to patients who needs blood in emergency and supports SDG Goal 3. It provides Individual and Organizations to host blood drives, request for blood and can search particular donor based on Name, Blood group, Address. Developed by KLE Technological students, it emphasizes Good Health and Well-being

Problem Statement

This project aims to develop a application for seamless communication between blood donors and hospitals, ensuring timely blood donations and distribution to patients who needs blood in emergency.

Objectives

To Ensure timely access and sufficient blood supplies for medical emergencies and transfusion needs.

Stakeholders:

KLE Wellness Center

4th Gate, Udyambag



SDG Mapping & PI

SDG GOAL 3

Good Health and Well-being for all

Target 3.8: Achieve universal health coverage, including access to safe, effective, quality, and affordable essential medicines and vaccines for all.

Indicator 3.8.1: Universal health coverage implies that all people receive the quality health services they need.

Design

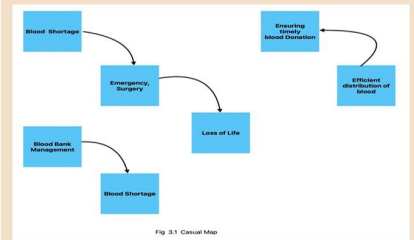
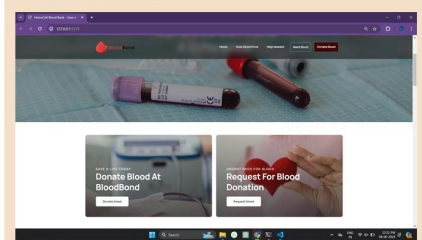


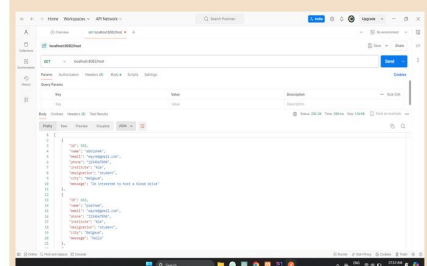
Fig 3.1 Casual Map

Implementation & Testing

Frontend:



Backend: (Postman API)



Test Case Table:

ID	Description	Expected Result	Actual Result	Result
DT01	Create new donate blood entry with all fields	The new donate blood entry is saved successfully and returned with an ID and the correct data fields.	The new donate blood entry is saved successfully and returned with an ID and the correct data fields.	Pass
DT02	Create new donate blood entry with missing field	An error indicating a NOT NULL constraint violation is returned.	An error indicating a NOT NULL constraint violation is not returned.	Fail
DT03	Read all donate blood entries	All existing donate blood entries are retrieved and displayed correctly.	All existing donate blood entries are retrieved and displayed correctly.	Pass
DT04	Delete a donate blood entry by ID	The specified donate blood entry is deleted successfully and is no longer present in the list of entries.	The specified donate blood entry is deleted successfully and is no longer present in the list of entries.	Pass